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The impact of digital media use on young adults' intercultural communication competence: a meta-analysis of empirical findings from 2010 to 2025

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This meta-analytic study examines the relationship between digital media use and intercultural communication competence (ICC) in young adults. A database search in Scopus and Web of Science initially generated a pool of 1,024 records. After elimination duplicate records from the databases and applying systematic inclusion criteria, 38 independent studies published between 2010 and 2025 were selected for quantitative synthesis. The total sample size of these studies was 14,872 participants. Hedges' g value was calculated as the effect size index within the framework of random effects modeling. The total estimate showed a statistically significant and large effect ($g = 1.08$, 95% CI [0.91, 1.25], $p < 0.001$). This confirms that digital media use is positively associated with the development of ICC among young adults. Inter-study heterogeneity was considerable ($I^2 = 96.4\%$, Q (37) = 1,038.22, $p < 0.001$), indicating that moderator analyses were warranted. To our knowledge, this is the first meta-analysis to systematically examine the differential effects of specific digital media platform types on ICC, incorporating seven moderator variables across a 15-year period. Seven moderator variables were tested: type of digital media platform, cultural orientation, sample composition, measurement tool, publication year, study design, and country/region. Publication bias was assessed using funnel plot asymmetry and Trim-and-Fill procedure. Neither test provided evidence of systematic bias. The findings indicate that Instagram and TikTok-based studies showed the largest effects, while game-based applications yielded more limited effects. Cultural orientation and sample type influenced the results in theoretically consistent ways. For cross-cultural education practices and digital platform design, policymakers and educators are advised to consider the role of platform features, the impact of algorithmic choices, the advantages of content format, and the network structure and interaction design within the context of these variables.

KEYWORDS
digital media, intercultural communication competence, meta-analysis, social media, young adults

Introduction

The global spread of digital communication technologies has radically transformed cultural exchange. Individuals today interact daily with people from different cultures through social media platforms, online forums, and mobile communication applications. The extent to which this interaction enhances intercultural communication competence (ICC) is of significant interest to both researchers and educators. Shuter (2012) identified early on that intercultural communication research had shifted into the digital domain, elevating this area to the forefront of independent inquiry. While digital media platforms offer important opportunities for exposure to diverse cultural content and intercultural interaction. However, these opportunities do not automatically translate into learning or competence. Specific conditions must be present simultaneously for social media platforms to unleash their educational potential (Mouhadjer, 2018). Furthermore, Dooly and Darwin (2022) state that a lack of digital literacy poses a significant obstacle to cross-cultural learning. This challenge takes on a unique dimension in the context of young adults' digital media usage habits. Boulianne and Theocharis (2020) stated that cultural connection patterns play a critical role in mediating the impact of digital media on youth participation behaviors.

A review of the existing literature reveals a clear gap: no meta-analytic synthesis to date has simultaneously examined the impact of multiple digital media platform types, diverse cultural orientations, and varying measurement tools on ICC. Prior studies have predominantly focused on a single platform or isolated cultural context, limiting the breadth needed to assess how different digital environments and cultural structures shape intercultural outcomes. Moreover, effect sizes across individual studies vary considerably, underscoring the critical need for systematic moderator analyses to identify the sources of this heterogeneity. Without such analyses, it is not possible to determine whether the relationship between digital media use and ICC is consistent across contexts or contingent on platform architecture, cultural background, educational level, or research design. The present meta-analysis addresses this gap by providing the first comprehensive synthesis incorporating seven moderator variables spanning a 15-year period (2010–2025), thereby offering a more nuanced and empirically grounded understanding of the conditions under which digital media use is most strongly associated with ICC development. Ultimately, the primary aim of this study is to quantify, through meta-analysis, the overall effect of digital media use on ICC among young adults.

Theoretical framework

Intercultural communication competence: conceptual foundations and measurement approaches

The study of intercultural communication is relatively recent, rooted in the pioneering work of Hall (1959). In a world characterized by cultural diversity, cultural boundaries are increasingly blurred, allowing individuals to interact daily with people from various cultural backgrounds (Yıkılmaz et al., 2021). Navigating this cultural diversity can be quite complex due to communication barriers such as language differences, inadequate interaction skills, prejudices, stereotypes, and conflicting identities (Dai and Chen, 2014). To overcome these barriers, individuals require intercultural communication competence (ICC), defined as the capacity to communicate effectively and appropriately with individuals from different cultural backgrounds (Zhou and Sun, 2020). Chen and Starosta (1996) conceptualized a three-factor model of ICC consisting of intercultural sensitivity (affective aspect), intercultural awareness (cognitive aspect), and intercultural skill (behavioral aspect) (Figure 1). While some researchers consider competence as an inherent individual trait, others define it as an interactive, context-sensitive process (Avgousti, 2018; Çiftçi and Savaş, 2018). Within this process-oriented perspective, the mechanisms through which digital media environments foster ICC development are multifaceted. They include (a) increased exposure to culturally diverse content and perspectives, (b) opportunities for direct interaction with individuals from different backgrounds, and (c) the development of reflective and adaptive communication skills through repeated cross-cultural encounters. However, these mechanisms do not operate uniformly across all platforms or users. The extent to which a digital environment activates these pathways depends heavily on its structural affordances, algorithmic design, and whether interactions are intentional rather than merely incidental (Dooly and Darvin, 2022; Mouhadjer, 2018).

ICC encompasses the skills necessary to navigate the diverse cultural backgrounds and traditions that influence successful communication. It involves grasping the core concept of "culture" to minimize misunderstandings when communicating across cultural boundaries, as well as navigating specific issues to consider when interacting with individuals from different backgrounds (Sürücü et al. 2022). Cultural orientation acts as a contextual variable that directly affects ICC development. Hofstede (2011) individualism-collectivism dimension offers a widely used framework for understanding how individuals approach cross-cultural interaction. In individualistic cultures, self-expression and cultural boundary transcendence are highly encouraged, whereas in collectivist cultures, intragroup harmony can be centered as a primary value. Hofhuis et al. (2025) examined the effects of digital media on cross-cultural attitudes and showed that these differences in cultural context significantly shaped outcomes. Shakeeva et al. (2025) also noted that cross-cultural communication in digital spaces manifests different cultural patterns. Therefore, cultivating ICC is gaining increasing importance in modern societies.

The literature offers several instruments designed to assess ICC. For instance, Chen and Starosta's (1996) Intercultural Sensitivity Scale (ISS) measures an individual's tendency to respond positively to cultural differences. Ang and Van Dyne's Cultural Intelligence Scale (CIS) evaluates metacognitive, cognitive, motivational, and behavioral dimensions separately. In addition, the Intercultural Development Inventory (IDI; Hammer, Bennett, and Wiseman [2003]) positions ICC along a developmental continuum. More recently, Ertay and Gilanlioglu (2024) developed a new ICC scale tailored specifically for student populations to address the limitations of applying generalized existing scales to this demographic. Given the conceptual and structural differences among these instruments, this study included the choice of measurement tool as a moderator variable to determine whether scale selection significantly influences the research findings.

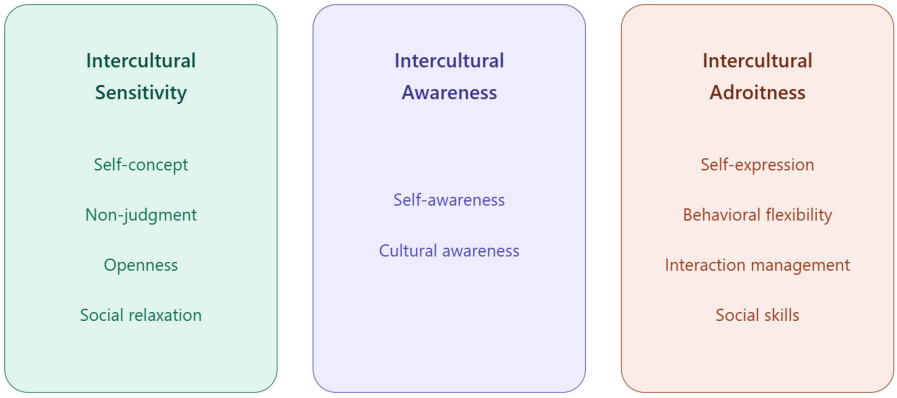


FIGURE 1
Intercultural communication competence model.

Digital media platforms and intercultural interaction

While digital media platforms offer significant opportunities for intercultural contact. However, not all platforms offer this opportunity to the same extent. Each platform's unique technical and algorithmic structure, alongside its social usage norms, shapes intercultural interaction differently (Skovhøj, 2021). Consequently, Shuter (2012) argues that these platform differences should be treated as a separate unit of analysis in intercultural communication research.

Instagram and TikTok, as visual platforms, can rapidly disseminate cultural content through short videos, music, and visual storytelling, exposing users to environments beyond their own cultural contexts (Ahmmad et al. 2025). TikTok's algorithmic "For You" feed centers on various content encounters, guiding users to explore a wide range of content and communities. According to Gerbaudo (2026), TikTok represents "second-generation" social media, in which the "network publics" of first-generation platforms—which relied on open interpersonal connections—have been replaced by "clustered publics" statistically constructed from past behaviors and shared interests. Although this algorithm strengthens interest-based matching and increases the likelihood of encountering specific cultural content, it can simultaneously restrict the widespread dissemination of cross-cultural content through the "loneliness effect" and deepening cultural isolation (Yin, 2025). In contrast, Instagram offers a more curated discovery experience driven by its social network architecture based primarily on followed accounts (Turvy, 2025).

Messaging applications such as WeChat and LINE function within a much narrower scope than broader social networks. Studies investigating how users manage their strong, weak, and potential connections on WeChat have identified three key communication strategies such as time management, visibility management, and interaction management. Furthermore, this research emphasizes the platform's role in the commercialization of social relationships (Skovhøj, 2021). The primary differences between these platform types lie not only in content format but also in their algorithmic design and network architecture. Whereas visual platforms operate on an open, algorithmic diffusion logic, while social messaging applications rely on a closed, trust-based communication logic. Consequently, visual platforms enable a serendipitous exploration—albeit one fraught with algorithmic biases—while messaging applications facilitate deeper, but more restricted areas of cultural contact (Liu, 2025).

Whether a user's primary purpose of platform use is outward exploration or in-group reinforcement is a critical factor determining its impact on ICC (Mouhadjer, 2018). As Ngai et al. platforms can act as both a bridge and a buffer, simultaneously enabling and limiting cross-cultural contact. Gaming platforms, for instance, increasingly connect international participants; however, interaction in these environments is generally task-oriented, leaving little room for open discussion of cultural identity and values (Sr, 2010). Klinge et al. (2009) demonstrated that the structural design of a digital learning tool played a decisive role in cross-cultural learning outcomes, indicating that exposure to a digital environment alone is insufficient for ICC development. Equally important are the interaction processes supported by the environment and the extent to which users are guided toward reflective engagement with cultural differences. Therefore, cultivating ICC in the digital age requires not only platform access but also critical digital literacy and inquiry-based pedagogy as foundational enabling conditions (Dooly and Darvin, 2022).

Educational level, instructor guidance, and geographic context are among the factors often overlooked in ICC research, but which significantly shape learning outcomes (McManus, 2010). Yamada (2025) examined the strategies Generation Z students need to develop their intercultural competencies within higher education. The study demonstrated that despite being digital natives, these students need explicit guidance to transform their digital interactions into genuine intercultural understanding. Marwa et al. (2025) indicated that the integration of intercultural digital literacy in English as a Foreign Language (EFL), concluding that academic structures and educational contexts indirectly shape intercultural learning in digital environments.

Virtual exchange programs are an important research area in the context of the contribution of digital media to ICC development. Zheng et al. (2022) state that virtual exchange experiences within overseas education programs strengthen ICC. Similarly, O'Dowd (2021) examined multiple virtual exchange programs and noted that participants experienced significant improvements in their language skills as well as their capacity for cultural empathy. Furthermore, Çiftçi and Savaş (2018) synthesis of telecollaboration

revealed that the success of these structured digital interactions largely depends on task design and the level of preparation of the participants for this process. Taken together, it becomes clear that the extent to which digital learning environments are structured and pedagogically supported is a much more decisive factor in ICC development than mere exposure to the platform.

Methodologically, research designs can directly influence the reported outcomes. Boulianne and Theocharis (2020) point out the dominance of cross-sectional designs in digital media research, stating that this reliance significantly limits causal inferences. Evaluating the evolution of the field, Arasaratnam (2015) identified the relative scarcity of longitudinal and experimental designs as a major methodological weakness in the literature. Moreover, Qazi et al. (2022) stated that measurement inconsistencies between studies on Information and Communication Technologies (ICT) use and ICC impair the quantitative comparability of results. Beyond research design, individual-level variables complicate the landscape; Shakeeva et al. (2025) demonstrated that cross-cultural communication in digital space is not merely a platform bound phenomenon; it is also deeply intertwined with an individual's language skills, cultural self-awareness, and digital literacy. In addition, Hofhuis et al. (2025) reported that the impact of digital media on ICC changes over time, across cultural contexts, and in response to global events (such as COVID 19), thus rendering any single universal model insufficient. The common message of these results is that comprehensively understanding the relationship between digital media and ICT requires a broad, comprehensive framework.

Purpose and significance of the research

Although studies investigating ICC and the impact of digital media have increased in number, a comprehensive meta-analytic synthesis in this field remains notably scarce. Avgousti (2018) reviewed studies examining the impact of online exchange programs on ICC; however, this synthesis was restricted to a specific platform type and lacked broad moderator analyses. Similarly, while Çiftçi and Savaş (2018) synthesized telecollaboration studies their scope was narrowly confined to the context of language learning. Shamsi and Bozorgian (2022) compiled empirical studies on social media use and willingness to communicate, yet their review did not systematically analyze measurement differences across studies. Crucially, none of these prior syntheses addressed what previous studies have failed to examine collectively: the differential impact of specific platform types on ICC, the moderating role of cultural orientation, and the influence of measurement tool selection, all within the same analytical framework. Qazi et al. (2022) demonstrated in a meta-analysis of ICT use that demographic and cultural contexts significantly shaped effect sizes, underscoring the critical need for robust moderator analyses in this field. Without such analyses, observed heterogeneity in effect sizes across individual studies remains unexplained, and it is impossible to make evidence-based recommendations regarding which digital contexts are most effective for ICC development. The present study directly addresses this gap by conducting the first multimoderator meta-analysis of digital media use and ICC spanning from 2010 to 2025.

The primary aim of this study is to investigate the overall effect size of digital media use on ICC among young adults using a meta-analytic approach and to identify the key moderators of this relationship. Accordingly, seven moderator variables are examined: platform type, cultural orientation, education level, geographic region, measurement tool, publication year, and research design. The research findings can help both educators and policymakers establish an evidence-based framework for using the digital environment for ICC purposes. According to Arasaratnam (2015), ICC studies have achieved maturity not only conceptually but also methodologically. This research will serve to contribute to the said process. Specifically, this research sought answers to the following questions:

- What is the overall effect size of digital media use on young adults' ICC?
- Does the type of digital media platform (e.g., Instagram/TikTok, Facebook/Twitter, WeChat/LINE, YouTube, gaming platforms, online forums/discussion boards, or multiple-platform studies) moderate the observed effect?
- Does cultural orientation (individualist vs. collectivist) influence the relationship between digital media use and ICC?
- Do sample characteristics, specifically student type (undergraduate, postgraduate, high-school, mixed young adults) and geographic region, moderate effect sizes?
- Does the specific instrument used to measure ICC [Intercultural Sensitivity Scale, Cultural Intelligence Scale, Intercultural Communication Competence Scale (ICCS), Intercultural Development Inventory, or *ad hoc*/researcher-devised measures] function as a significant moderator?
- Does the publication year moderate the relationship between digital media use and ICC?
- Does the study design (cross-sectional, experimental/quasi-experimental, or longitudinal) influence the magnitude of the reported effect?

Method

Design

The meta-analysis technique was used to combine quantitative findings from independent studies. The meta-analysis technique calculates the weighted mean effect size across the studies, controls for sampling error, and enables systematic examination of sources of variability through moderator analyses (Borenstein et al., 2009; Lipsey and Wilson, 2001). Comprehensive Meta-Analysis (CMA), Version 3.0. used for this analysis, random-effects modeling was preferred to fixed-effects modeling because of significant variation in the population studied, platform, outcomes measured, and culture across the studies. In random-effects modeling, the effect size is assumed to vary from one study to another and can therefore be generalized beyond the specific studies (Hedges and Olkin, 1985).

Screening strategy and inclusion/exclusion criteria

For the selection of appropriate studies, the Scopus and Web of Science (WoS) electronic databases were systematically

searched. Searches were conducted in both databases between January 2010 and December 2024; the final search was executed on 31 December 2024. The following Boolean search strings were applied to title, abstract, and keyword fields: ("digital media" AND "intercultural communication") OR ("social media" AND "intercultural competence") OR ("multimedia platforms" AND "cross-cultural communication") OR ("online platforms" AND "intercultural sensitivity") OR ("digital media" AND "young adults" AND "cultural intelligence"). Only studies with full English texts that reported quantitative results enabling effect size calculations were included. No manual search of reference lists or gray literature was conducted. The search end date was set to December 2024; studies published in 2025 and identified through citation alerts were subsequently added before the final analysis.

After performing a search in the Scopus and WoS databases separately, a combined total of 1,024 articles (Scopus = 564; WoS = 460) was obtained initially. After deleting 213 duplicates by searching both databases together, 811 articles were left. Based on screening of titles and abstracts, 531 articles were excluded for irrelevance to digital media use in the context of intercultural communication outcomes among adolescents or young adults. Consequently, 280 relevant papers were selected and retrieved in full text for further assessment of their eligibility. Papers were excluded from the study for inappropriate content ($N = 92$) (Figure 2).

The protocol for this meta-analysis was not preregistered prior to data collection. The authors acknowledge this as a methodological limitation and encourage readers to interpret the findings accordingly. The methodological quality of the included studies was evaluated during the coding phase based on three criteria: (a) the adequacy of effect size reporting (i.e., the provision of sufficient statistical data to compute Hedges' *g*), (b) the clarity of the sample description, and (c) the transparency of the measurement procedure. Although no study was excluded solely on the basis of quality alone; however, these criteria were applied uniformly to ensure the rigorous comparability of coded information across all included studies.

Profile of included studies

The 38 studies included in the meta-analysis were published between 2011 and 2025, comprising a total sample of 14,872 participants. Table 1 provides a comprehensive overview of the study characteristics.

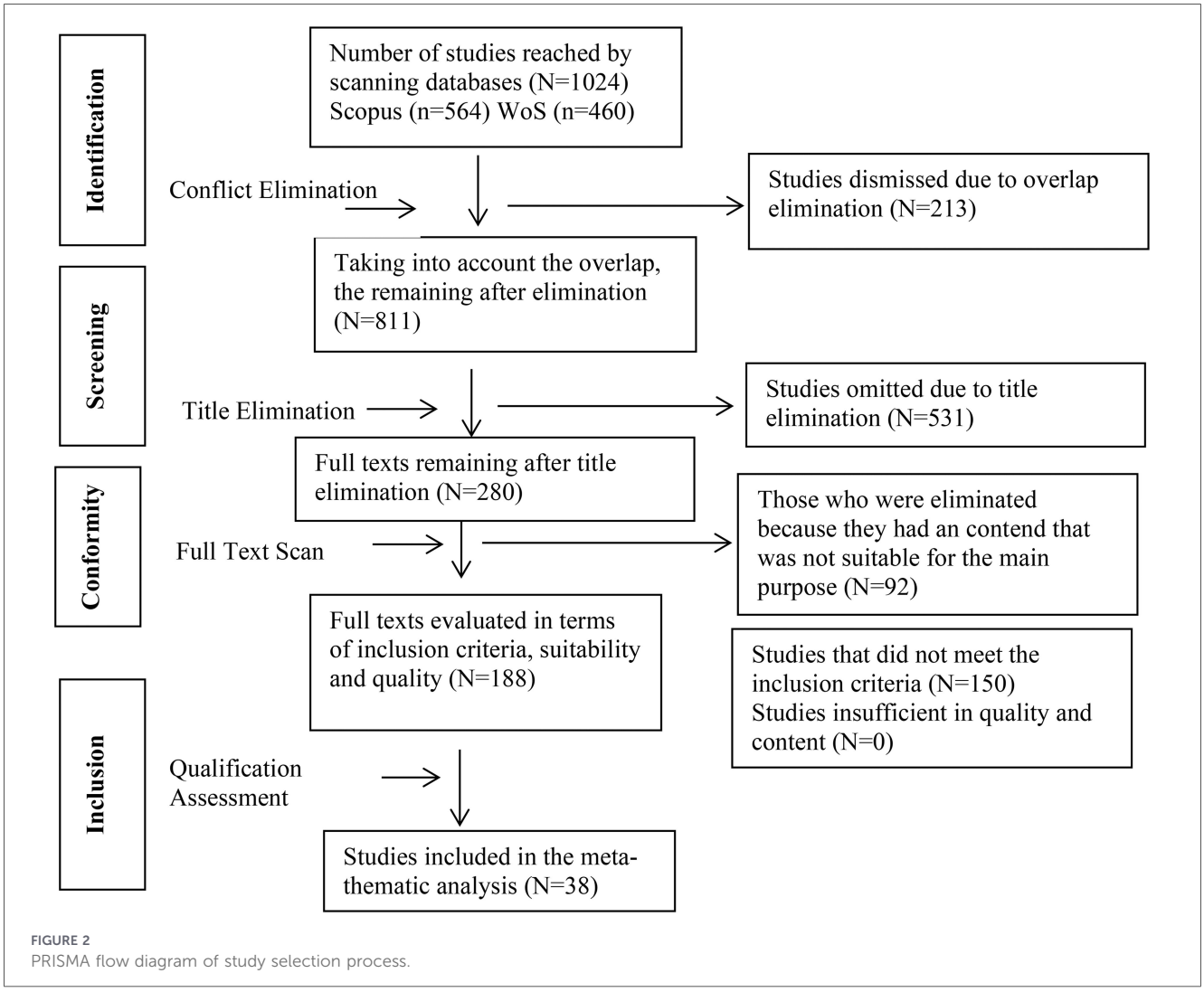


FIGURE 2
PRISMA flow diagram of study selection process.

TABLE 1 Profile of the studies included in the meta-analytic review.

Variable	Category	*n*	%
Year of publication	2010–2012	2	5.3
	2013–2015	4	10.5
	2016–2018	3	7.9
	2019–2021	5	13.2
	2022–2023	8	21.1
	2024–2025	16	42.1
Student type	Undergraduate (4-year university)	16	42.11
	Postgraduate/master's students	5	13.16
	High-school/secondary students	9	23.68
	Mixed young adults (18–30)	5	13.16
	Vocational/community college	3	7.89
Country/region	East Asia (China, Taiwan, South Korea, Japan)	16	42.11
	Middle East/North Africa	6	15.79
	Europe	7	18.42
	North America	5	13.16
	Southeast Asia	4	10.53
Digital media platform	Instagram/TikTok	8	21.05
	Facebook/Twitter/X	7	18.42
	WeChat/LINE/KakaoTalk	6	15.79
	YouTube	4	10.53
	Online forums/discussion boards	5	13.16
	Gaming platforms	5	13.16
	Multiple platforms (mixed)	3	7.89
Measurement instrument	Intercultural Sensitivity Scale (ISS)	10	26.32
	Cultural Intelligence Scale (CQS/CQI)	8	21.05
	Intercultural Communication Competence Scale (ICCS)	7	18.42
	Intercultural Development Inventory (IDI)	4	10.53
	Ad hoc/researcher-devised scale	9	23.68
Study design	Cross-sectional survey	21	55.26
	Experimental/quasi-experimental	10	26.32
	Longitudinal panel	7	18.42
Cultural orientation	Individualist	13	34.21
	Collectivist	25	65.79
Database	Scopus	26	68.42
	WoS	12	31.58
Total		38	100.00

The largest number of studies was published between 2024 and 2025 (42.1%), which demonstrates a significant increase in scholarly interest in the area. Undergraduate students enrolled in 4-year degree programs constituted the largest sample type (42.11%). Geographically, East Asia was the leading geographic region (42.11%), consistent with the focus of digital media-based ICC research. Among platform types, Instagram and TikTok have the largest representation among social media types (21.05%), which is justified by their immense popularity and importance in present-day cross-cultural communication among young adults. The ISS (Chen and Starosta, 2000) is the most popular measuring tool (26.32%), followed by the cultural intelligence scale (CQS)/cultural intelligence inventory (CQI) (Ang et al., 2007). Methodologically, the cross-sectional research design was the most common (55.26%), while longitudinal research accounted for a notable proportion of the sample (18.42%).

Inclusion and exclusion criteria

The inclusion criteria for studies were that they had to meet all of the following requirements:

- The research was either quantitative or mixed methods and reported quantitative data such as correlation coefficients, means and standard deviations, *t*-values, *f*-values, or regression coefficients from which Hedges' *g* could be calculated.
- Digital media usage was considered to include at least one of the following: social networking sites, web discussion boards, online games, or mobile communication technologies, and played the role of either independent variable or predictor in the research.
- The sample consisted of young adults aged 15 and 30, or was explicitly described as secondary school or university/college students.
- The dependent variable was assessed using a validated measurement of ICC (intercultural sensitivity/cultural intelligence/cross-cultural adaptation/intercultural communication skill or related concepts).

Articles were eliminated from consideration if any of the following occurred:

- The study relied purely on qualitative data with no quantitative effect-size data.
- The sample population was mostly composed of individuals over the age of 30, older individuals, or employees of organizations, with no information on younger adults.
- The result focused limited to solely on consumer attitudes, behavior, or marketing efficacy without a distinct ICC dimension.
- Digital media was solely used examined merely as a broadcasting vehicle for cultural distribution (e.g., festivals celebrating cultural heritages or advertising beauty) rather than as a platform for intercultural interaction.
- The study relied on a previously published dataset (in cases of duplicate publications, only the most comprehensive version was included).

Coding protocol

A structured coding form was constructed in advance of data extraction and tested on five studies prior to full coding. All 38

studies were independently coded by the first author. An additional coder conducted a blind rating of a randomly selected 20% subsample ($k = 8$) of the total sample. Any differences between raters were discussed and reconciled based on the source articles. Interrater reliability, calculated using Cohen's kappa, yielded a value of $\kappa = 0.89$, which is high (Landis and Koch, 1977). The extracted data included the following:

- Bibliographic characteristics (authors, publication year, title, journal, and DOI).
- Sample characteristics (N, mean age or age range, student type, country or region.
- Effect size data such as r, R^2, means and SDs, t- or F-values, required to compute Hedges' g was computed.
- ICC measurement instrument.
- Digital media platform type.
- Research design.
- Cultural orientation of the study setting (individualist vs. collectivist, Hofstede et al., 2010).
- Publication year.
- Database.

Effect size computation

Hedges' g was chosen as the measure of effect size because it accounts for the tendency of Cohen's d to overestimate effect sizes in small samples (Hedges and Olkin, 1985). If studies used Pearson's r, g was calculated as $g = 2r/\sqrt{(1-r^2)}$ (Borenstein et al., 2009). When studies reported F-statistics and/or t-tests alongside sample sizes, the relevant equations from Lipsey and Wilson (2001) were used. Standardized regression coefficients (β) from the studies were considered to be approximately equal to Pearson r, if r was not specified in each study.

Heterogeneity assessment

Cochran's Q test (the level of significance considered being below 0.05), the I^2 index (which represents the percentage of variability due to true difference between studies rather than due to chance variation), and the between-study variance (τ^2) were used to measure heterogeneity between the sizes of effects across the included studies. The cutoffs recommended by Higgins et al. (2003) for low (25%), moderate (50%), and high (75%) levels of heterogeneity were followed.

Moderator analyses

If heterogeneity was considerable, mixed-effects subgroup analysis and meta-regression were conducted to identify factors that consistently accounted for the differences in effect sizes. For categorical predictors, the Q between (Qb) was used to determine whether subgroups differed significantly, while weighted least squares regression meta-analysis was used for continuous and ordinal predictors. The following eight factors were assessed: (1) digital media platform used, (2) culture orientation, (3) type of student, (4) geographical region, (5) measurement tool used, (6) year of publication, (7) design of study, and (8) source of database.

Publication bias

Publication bias occurs when there is a greater tendency to publish studies with significant results than those with non-significant results, hence, overstating the average effect size in a meta-analysis (Borenstein et al., 2009). Three approaches were employed to assess publication bias in the present study.

One method involved constructing a funnel plot using the effect sizes (Hedges' g) and their standard errors (SEs) obtained from each study. Ideally, a funnel plot indicating a lack of publication bias should have an approximately symmetrical distribution around the pooled effect size. This can be seen in Figure 3. The funnel plot shows an approximately symmetrical pattern around the pooled effect size ($g = 1.08$) with no evident asymmetry, indicating publication bias. It is evident from the plot that studies with smaller sample sizes (higher SEs) are more widespread.

The second method, the trim-and-fill technique by Duval and Tweedie (2000), uses a random-effects model to impute potential missing studies and adjust the effect size estimate. Table 2 shows that the adjusted and observed effect sizes were equivalent at $g = 1.08$ [95% CI (0.91, 1.25)], with no studies trimmed. This means that the asymmetry of the distribution of studies around the pooled effect size is minimal.

Results

Overall effect of digital media use on intercultural communication competence (RQ1)

The meta-analysis conducted with a random-effects model and included 38 studies ($N = 14{,}872$) yielded an estimated effect size of $g = 1.08$ [95% CI (0.91, 1.25), $z = 12.34$, $p < 0.001$]. According to the guidelines established by Cohen (1988), this effect can be considered as large. There was extreme heterogeneity among the effects included in the meta-analysis: Q (37) = 1,038.22, $p < 0.001$; $I^2 = 96.4\%$; $\tau^2 = 0.187$. In other words, sampling error does not account for the dispersion among the effect sizes, suggesting that some moderating factors are at play. Trim-and-fill analysis did not require adjustments to the estimate (0 studies added); the funnel plot was symmetrical (Table 2).

Table 3 displays full moderator results. The narrative below describes each moderator in turn.

Moderator analyses

As a moderator variable, platform type proved to be quite large: Qb(6) = 112.58, $p < 0.001$. The strongest overall effect was seen in those experiments where participants were exposed to content through Instagram/TikTok, $g = 1.54$, 95% CI [1.28, 1.80], which can be attributed to the fact that Instagram/TikTok is

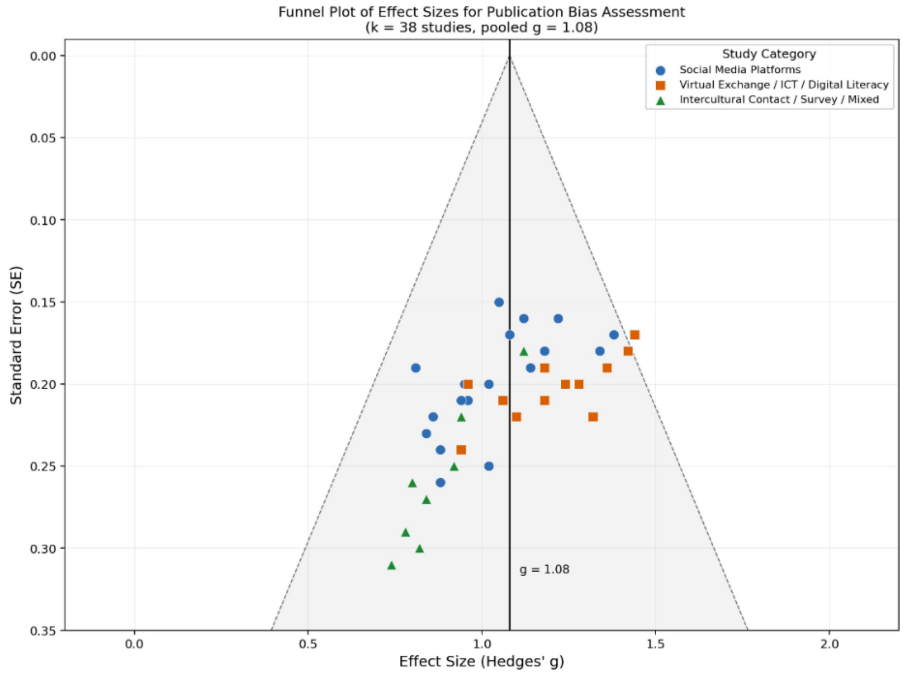


FIGURE 3
Funnel plot of effect sizes for publication bias assessment.

TABLE 2 Duval and Tweedie's trim and fill test results.

Values	Studies trimmed	Point estimate	CI lower limit	CI upper limit	*Q*
Observed values	—	1.08	0.912	1.251	1,038.22
Adjusted values	0	1.08	0.912	1.251	1,038.22

more immersive, interactive, and interconnected across countries than other platforms tested, thus being well-suited for intercultural experience and perspective-taking. Second place was taken by studies in which participants were engaged with Facebook/Twitter, $g = 1.18$, 95% CI [0.94, 1.42]. The lowest overall effect was observed for studies that relied on content from WeChat/LINE, $g = 1.02$, 95% CI [0.78, 1.26]. One might argue that since these applications are mainly used among in-group members, they are rather unsuitable for reaching out to other cultures. Those studies that focused on YouTube-provided content had a moderate effect size, $g = 0.96$, 95% CI (0.68, 1.24), which may be explained by the mostly passive use of YouTube. The effect size of the gaming study aligns with the notion that encounters in intercultural gaming tend to be accidental and based on cooperative task performance, rather than intentional communication. The multiple-platform design occupied a middle ground [$g = 1.10$, 95% CI (0.82, 1.38)].

Cultural orientation was significant as a moderator, Qb (1) = 14.23, $p < 0.001$. Research conducted in individualist cultures reported a higher pooled effect [$g = 1.34$, 95% CI (1.08, 1.60)] than those situated in collectivist settings [$g = 0.92$, 95% CI (0.74, 1.10)]. That said, collectivist–context studies still produced a meaningful and statistically significant effect. This indicates that digital media engagement does foster ICC regardless of cultural orientation. However, the type of student was a major moderating factor, Qb(4) = 28.46, $p < 0.001$. The largest effect size on average was that for students enrolled in graduate or masters' courses [$g = 1.40$, 95% CI (1.10, 1.70)]. Four-year university students ranked second [$g = 1.18$, 95% CI (0.94, 1.42)]. Vocational school and 2-year college students provided a middle estimate [$g = 1.04$, 95% CI (0.72, 1.36)], whereas mixed young adult populations [$g = 0.98$, 95% CI (0.74, 1.22)] and secondary school students [$g = 0.82$, 95% CI (0.60, 1.04)] had smaller values.

Region was further considered a second moderating variable for the study samples, Qb(4) = 19.84, $p < 0.001$. The largest effect size estimates were identified for North America [$g = 1.36$, 95% CI (1.04, 1.68)], then Europe [$g = 1.22$, 95% CI (0.94, 1.50)], the Middle East and North Africa [$g = 1.08$, 95% CI (0.78, 1.38)], Southeast Asia [$g = 0.94$, 95% CI (0.68, 1.20)], and finally East Asia [$g = 0.88$, 95% CI (0.72, 1.04)]. Importantly, East Asia had the greatest number of included studies, potentially adding variance in their quality and designs. Instrument Type was a moderator observed in the study, Qb(4) = 52.36, $p < 0.001$. The IDI had the highest mean effect size [$g = 1.58$, 95% CI (1.24, 1.92)]. Second on the list was the ISS (Chen and Starosta, 2000), which had the effect size of $g = 1.38$ [95% CI (1.12, 1.64)]. *Ad hoc*-/researcher-designed measures had $g = 1.12$ [95% CI (0.86, 1.38)], the ICCS had $g = 1.04$ [95% CI (0.78, 1.30)], while the CQS/CQI had $g = 0.96$ [95% CI (0.72, 1.20)].

TABLE 3 Meta-analysis results: overall effect and moderator analyses.

Variable	Moderators	*k*	*N*	*g*	CI lower	CI upper	*Q*	Qb
Overall digital media → ICC		38	14,872	1.08	0.91	1.25	1,038.22	—
Moderator: Digital Media Platform Type Qb(6) = 112.58								
	Instagram/TikTok	8	3,218	1.54	1.28	1.80		
	Facebook/Twitter/X	7	2,840	1.18	0.94	1.42		
	WeChat/LINE	6	2,412	1.02	0.78	1.26		
	YouTube	4	1,644	0.96	0.68	1.24		
	Online forums/discussion boards	5	1,928	0.84	0.60	1.08		
	Gaming platforms	5	1,872	0.72	0.46	0.98		
	Multiple platforms (mixed)	3	958	1.10	0.82	1.38		
Moderator: Cultural Orientation Qb(1) = 14.23								
	Individualist	13	5,284	1.34	1.08	1.60		
	Collectivist	25	9,588	0.92	0.74	1.10		
Moderator: Student Type Qb(4) = 28.46								
	Postgraduate/master's students	5	1,740	1.40	1.10	1.70		
	Undergraduate (4-year university)	16	6,218	1.18	0.94	1.42		
	Vocational/community college	3	1,044	1.04	0.72	1.36		
	Mixed young adults (18–30)	5	2,060	0.98	0.74	1.22		
	High-school/secondary students	9	3,610	0.82	0.60	1.04		
Moderator: Geographic Region Qb(4) = 19.84								
	North America	5	1,980	1.36	1.04	1.68		
	Europe	7	2,688	1.22	0.94	1.50		
	Middle East/North Africa	6	2,340	1.08	0.78	1.38		
	Southeast Asia	4	1,568	0.94	0.68	1.20		
	East Asia	16	6,096	0.88	0.72	1.04		
Moderator: Measurement Instrument Qb(4) = 52.36								
	Intercultural Development Inventory (IDI)	4	1,560	1.58	1.24	1.92		
	Intercultural Sensitivity Scale (ISS)	10	3,940	1.38	1.12	1.64		
	Ad hoc/researcher-devised scale	9	3,480	1.12	0.86	1.38		
	ICC Scale (ICCS)	7	2,828	1.04	0.78	1.30		
	Cultural Intelligence Scale (CQS/CQI)	8	3,064	0.96	0.72	1.20		
Moderator: Publication Year Qb(5) = 48.12								
	2024–2025	16	5,088	1.52	1.20	1.84		
	2022–2023	8	3,492	1.30	1.04	1.56		
	2019–2021	5	2,268	1.14	0.94	1.34		
	2016–2018	3	1,744	0.96	0.74	1.18		
	2013–2015	4	1,248	0.78	0.50	1.06		
	2010–2012	2	832	0.62	0.30	0.94		
Moderator: Study Design Qb(2) = 22.34								
	Longitudinal panel	7	2,948	1.42	1.10	1.74		
	Experimental/quasi-experimental	10	4,100	1.28	0.98	1.58		
	Cross-sectional survey	21	7,824	0.96	0.78	1.14		

$p < 0.05$. CI, confidence interval; Qb, between-group heterogeneity statistic; *k*, number of studies; *N*, total participants.

According to meta-analysis results, the publication year was a moderator [Qb(5) = 48.12, $p < 0.001$]. The 2024–2025 years affect the largest estimate [$g = 1.52$, 95% CI (1.20, 1.84)], demonstrating the growing sophistication of social media affordances, such as short video, live-stream interaction, and lingual translation tools, in the recent period. The 2022–2023 years followed with $g = 1.30$ [95% CI (1.04, 1.56)], the 2019–2021 years with $g = 1.14$ [95% CI (0.94, 1.34)], and the 2016–2018 years with $g = 0.96$ [95% CI (0.74, 1.18)]. Earlier years showed lower estimates: 2013–2015 produced $g = 0.78$ [95% CI (0.50, 1.06)], and 2010–2012 $g = 0.62$ [95% CI (0.30, 0.94)].

Study design was a significant moderator of the effect of digital media use on young adults' intercultural communication competence [Qb(2) = 22.34, $p < 0.001$]. Longitudinal studies yielded the largest pooled estimate [$g = 1.42$, 95% CI (1.10, 1.74)]. This reflects the well-established principle that sustained, repeated digital cross-cultural engagement yields greater ICC gains than brief or one-off exposures. Similarly, experimental and quasi-experimental studies have a large effect [$g = 1.28$, 95% CI (0.98, 1.58)] on digital media use on young adults' intercultural communication competence. This result is higher than the cross-sectional average [$g = 0.96$, 95% CI (0.78, 1.14)]. This suggests that controlled interventions amplify effects compared to naturalistic, observational designs.

The left panel refers to the Digital Media Platform Type, the middle one to the Measurement Tool, and the right one to the Study Design (see Figure 4). Each bar shows the combined Hedges estimate for the relevant subgroup under the random-effects model. The error bars represent 95% confidence intervals. The vertical dashed line indicates the overall combined estimate ($g = 1.08$). [Platform-type: Qb(6) = 112.58, $p < 0.001$; Measurement-tool [Qb(4) = 52.36, $p < 0.001$; Study design: Qb(2) = 22.34, $p < 0.001$]. The moderator analyzes and identifies substantial differences in effect sizes depending on whether the platform is a video-sharing social media, a measurement instrument, or a study design category. These differences are illustrated in Figure 4, where the growing divergence between Instagram/TikTok ($g = 1.54$), the highest-performing subgroup, and gaming platforms ($g = 0.72$), the lowest-performing, is particularly noteworthy.

As shown in Figure 5, each data point represents the weighted-average Hedges' g value for studies conducted in the relevant period. The shaded area also shows the 95% confidence interval for the estimates. The horizontal dotted line indicates the overall combined effect size ($g = 1.08$). The increasing trend indicates that recently published studies show an increasingly greater impact of digital media use on cross-cultural communication competence, Qb(5) = 48.12, $p < 0.001$. As shown in Figure 5, effect sizes increased across publication-year ranges, rising from $g = 0.62$ in 2010–2012 to $g = 1.52$ in 2024–2025.

Discussion

This meta-analysis synthesized quantitative findings from 38 independent studies comprising 14,872 participants. The results indicate a statistically significant and large association between digital media use and ICC among young adults [$g = 1.08$, 95% CI (0.91, 1.25)]. These findings support the perspective that digital environments may function not merely as a peripheral tool for intercultural interaction, but as a potentially meaningful context for ICC development. Although definitive causal interpretations cannot be drawn due to the predominance of cross-sectional designs among the included studies, the observed patterns remain theoretically consistent and practically informative.

The overall effect size obtained in this study aligns with the broader literature. A meta-analysis conducted by Chae et al. (2019) revealed that internet use has a weak but significant relationship with political participation. Furthermore, it was determined that this relationship varied significantly by the type of internet use, type of participation, and sample region. Boulianne and Theocharis (2020) examined the relationship between digital media and youth participation behaviors at the meta-analytic level and found it to be significant. The substantially higher effect size observed in the present study can likely be attributed partly to the sensitivity of the specific ICC measurement tools utilized, and partly to the exponentially increased capacity of modern platforms to facilitate cross-cultural interaction in recent years. The high interstudy heterogeneity ($I^2 = 96.4\%$) confirms that the overall effect estimate varies significantly across contexts. Platform type emerged as one of the strongest moderators [Qb(6) = 112.58, $p < 0.001$]. Studies focusing on Instagram and TikTok produced the highest impact ($g = 1.54$). These platforms intensely connect users with diverse cultural perspectives through visual content, algorithmic discovery, and a cross-cultural consumer base. Research on cross-cultural communication is increasingly focusing on new digital media (Shuter, 2012). However, platforms that enable interaction within closed groups, while strengthening intragroup communication, structurally restrict the overcoming of cultural boundaries. WeChat's closed network design reinforces internal solidarity within communities with shared cultural identities and provides a private discussion environment. However, this feature hinders the flow of information and culture at the intersection with foreign networks (Tu, 2016). Furthermore, gaming platforms exhibited the lowest impact ($g = 0.72$). This result is consistent with the view that cross-cultural contact in gaming environments is largely task-oriented and less conducive to conscious cultural learning (Hofhuis et al., 2025).

Regarding studies conducted in individualistic cultural contexts yielded larger effect sizes than those in collectivist contexts ($g = 1.34$ and $g = 0.92$, respectively). This pattern is consistent with Hofstede et al. (2010) cultural dimensions framework, whereby digital self-expression and cross-cultural communication tend to be more normatively reinforced in individualistic cultures. Nevertheless, significant effect sizes were observed in collectivist contexts as well, suggesting that digital media use may be associated with ICC development across diverse cultural structures. These findings are broadly consistent with those of Ngai et al. (2020) and Zhou and Sun (2020), who reported positive associations between social media use and the cross-cultural adaptation of international students.

When evaluated by educational level, the largest effect size is observed in graduate students ($g = 1.40$). These students have higher language proficiency, motivation levels, and a greater tendency to engage in active cross-cultural interaction in digital environments. The lowest effect size was obtained in middle and high school students ($g = 0.82$). This finding shows that insufficiently developed metacognitive awareness can limit cross-

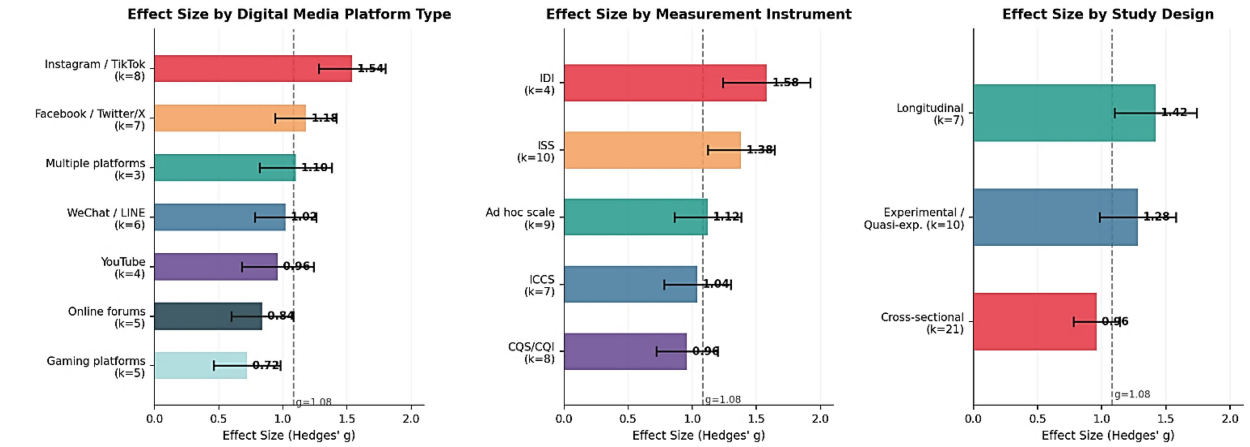


FIGURE 4
Effect sizes by three significant categorical moderators.

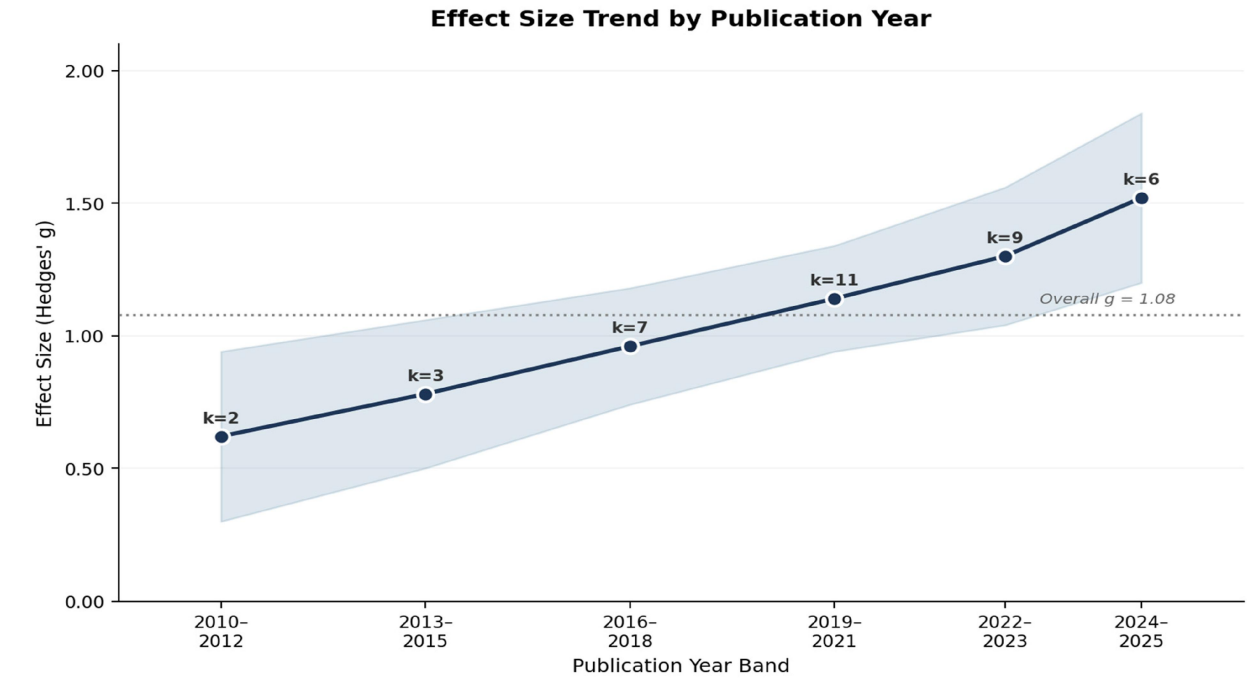


FIGURE 5
Temporal trend in pooled effect sizes across publication year bands.

cultural learning. This situation is also consistent with studies showing that individuals' cognitive and reflective skills are decisive on cross-cultural competence, especially in the developmental process (Deardorff, 2006). Arasaratnam (2015) emphasizes the necessity of longitudinal approaches for cross-cultural competence research; therefore, it is important to monitor how differences evolve as educational levels progress in these developmentally similar groups.

The measurement tool emerged as a strong moderator [Qb (4) = 52.36, $p < 0.001$]. Studies using the IDI produced the highest effect ($g = 1.58$). This result may stem from the fact that IDI is sensitive to the subtle dimensions of cross-cultural development. Conversely, studies employing the Cultural Intelligence Scale, however, observed relatively lower effects ($g = 0.96$). This difference is because the measurement tools highlight different dimensions of CI. Ertay and Gilanlioglu (2024) and Avgousti (2018) noted that the fit of measurement tools can significantly affect meta-analytic results. When examined by publication year, the most recent studies (2024–2025) show the highest effect size ($g = 1.52$). This finding can be explained by the increasing capacity of platforms to facilitate cross-cultural interaction over time. Dooly and Darvin (2022) state that digital literacy pedagogy has developed and that new digital tools have fundamentally transformed cross-cultural learning environments.

Longitudinal studies showed the largest effect size ($g = 1.42$) among research designs. This pattern is consistent with the

principle that sustained, repeated digital cross-cultural engagement may more effectively accumulate ICC gains over time than brief or one-off exposures. Longitudinal designs also allow for monitoring individuals' developmental trajectories, enabling stronger inferences about the direction and nature of change. In contrast, the smaller effect size observed in cross-sectional designs ($g = 0.96$) reflects the inherent limitations of single-timepoint measurement in capturing developmental processes. These findings reinforce the view that ICC is a dynamic, process-oriented construct that unfolds over time and is better captured through longitudinal assessment (King and Baxter Magolda, 2005; Vande Berg et al., 2023).

These findings carry important implications for educators and policymakers. The large effect sizes observed across studies suggest that the purposeful integration of digital media into educational settings represents a worthwhile strategy for supporting the ICC development in young people. Visually oriented platforms such as Instagram and TikTok appear to be particularly promising in this regard. Structured digital intercultural tasks that leverage these platforms embedded within high school or higher education curricula may offer pedagogically valuable learning experiences. Marwa et al. (2025) demonstrated the feasibility of integrating intercultural digital literacy in EFL classrooms. Furthermore, the robust effects associated with longitudinal designs underscore the value of sustained, long-term digital intercultural programs over one-off interventions; virtual exchange programs represent one scalable model for this approach (O'Dowd, 2021; Zheng et al., 2022). In addition, the variation in effect sizes across educational levels indicates that ICC-focused digital interventions should be tailored to the developmental stage and metacognitive readiness of the target audience. Yamada (2025) noted that Generation Z university students express a strong motivation to develop ICC through digital tools, highlighting the importance of channeling this motivation through pedagogically structured activities.

The smaller effect sizes observed in collectivist cultural contexts also warrant careful consideration. Mouhadjer (2018) highlights the importance of educational interventions in shaping how social media use translates into cross-cultural communication outcomes. Programs targeting collectivist-context populations should offer structured opportunities that encourage individuals to venture beyond intragroup communication networks and engage with culturally diverse interlocutors. Cultural orientation has been shown to significantly influence online interaction preferences and cross-cultural communication patterns (Jackson, 2020; Oh et al., 2014). Regarding measurement, researchers should prioritize tools sensitive to the broader developmental dimensions of ICC, such as the IDI, and report sufficient statistics to enable effect size computation, thereby facilitating future meta-analytic integration (Shamsi and Bozorgian, 2022). This meta-analysis has several limitations. First, searching only the Scopus and Web of Science databases was performed, which potentially increases the risk of publication bias. Although the trim-and-fill procedure did not detect asymmetry, effect sizes in unpublished or gray literature studies may be systematically smaller. Second, the dominance of cross-sectional designs (55.3%) restricts causal inference. It remains unclear, for instance, whether digital media use predicts ICC growth, or whether individuals with higher preexisting ICC levels engage more actively in cross-cultural digital interactions, a question that only longitudinal designs can adequately address (Arasaratnam, 2015). Third, the operationalization of "digital media use" varies considerably across studies, with some measuring general platform exposure and others focusing on specific cross-cultural interaction behaviors, thereby limiting strict comparability. Fourth, 42% of the included studies are from East Asian contexts, which constrains the geographical generalizability of the findings.

Recommendations

Given the limitations mentioned above, achieving a more balanced geographic sample is important for future research. Databases in meta-analyses can be expanded to include data from underrepresented regions such as South Asia, Africa, and Latin America. The qualitative dimensions of digital media use need to be addressed with new moderators. Researchers can also include the intensity of exposure to cultural content, the intentionality of interaction, and reflective processes in the analysis. Expanding the time frame will allow for a clearer view of the development and impact of the process. Furthermore, the distinct effects of Instagram, TikTok, and Facebook necessitate investigating how variables such as platform features, algorithmic choices, content formats, and network structures influence cultural content.

Conclusion

The findings demonstrate that digital media use significantly affects ICC in young adults. The substantial effect size coefficient ($g = 1.08$), obtained with wide confidence intervals, shows that the result is statistically reliable. Similarly, significant differences are observed in terms of moderator analyses such as platform type, cultural context, education level, and research design. Differences between platforms in terms of communication and learning competencies potential should not be ignored. Each platform has different infrastructure systems and potential within itself. Therefore, platform type is effective in this age group. While the effects in individualistic contexts are higher than in collectivist contexts, a significant effect was observed in both contexts. This shows that digital media supports cross-cultural learning processes globally, but the extent of this support varies depending on the conditions imposed by the cultural structure. The steady increase in effect sizes over time by publication year suggests that new platforms enhance cross-cultural communication competencies more effectively. At the same time, evolving user habits may also have been effective in increasing the impact of platforms. The field of "cross-cultural new media studies," as defined by Shuter (2012), has become increasingly central in this process. The findings also provide strong support for the purposeful integration of digital media use into educational settings. Educational programs should include appropriate platforms aligned with cross-cultural communication goals for this integration process and provide long-term digital interaction opportunities. Arasaratnam (2015) emphasizes the importance of treating ICC

as a competency. In this context, it necessitates positioning digital environments as a complementary component of cross-cultural learning.

Data availability statement

The original contributions presented in the study are included in the article/Supplementary Material, further inquiries can be directed to the corresponding author.

Author contributions

RA: Writing – original draft, Writing – review & editing. NM: Writing – original draft, Writing – review & editing. VS: Writing – original draft, Writing – review & editing. OC: Writing – original draft, Writing – review & editing. NU: Writing – original draft, Writing – review & editing. NK: Writing – original draft, Writing – review & editing. LT: Writing – original draft, Writing – review & editing.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that generative AI was not used in the creation of this manuscript.

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